# Fresnel Scattering with SCUFF-SCATTER-PERIODIC

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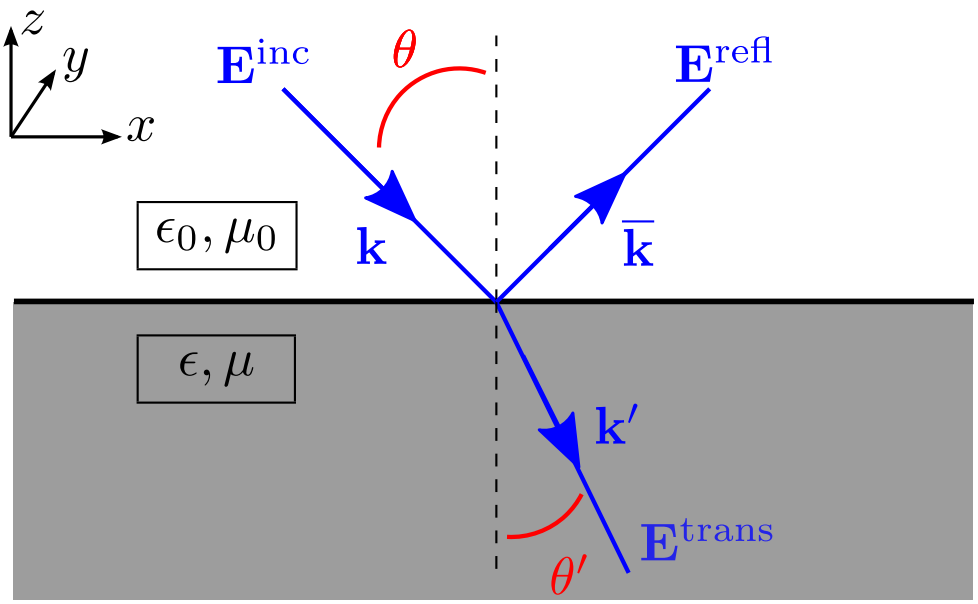



Figure 1: Fresnel scattering.

# 1 Exact Solution

Consider a plane wave at frequency $\omega$ impinging on an infinite planar material surface at incident angle $\theta$ (Figure 1).

## Incident field

In what follows, the superscripts "perp" and "par" describe the two possible polarizations: **E**−field perpendicular or parallel to the plane of incidence.

$$\mathbf{E}^{\rm inc,perp}(\mathbf{x}) = E_0 e^{i\mathbf{k}\cdot\mathbf{x}}\,\hat{\boldsymbol{y}}$$
$$\mathbf{E}^{\rm inc,par}(\mathbf{x}) = E_0 e^{i\mathbf{k}\cdot\mathbf{x}}\Big(\cos\theta\,\hat{\boldsymbol{x}} + \sin\theta\,\hat{\boldsymbol{z}}\Big)$$

where

$$\mathbf{k} = k\Big(\sin\theta\,\hat{\boldsymbol{x}} - \cos\theta\,\hat{\boldsymbol{z}}\Big), \qquad k = \frac{\omega}{c}.$$

## Bloch wavevector

The Bloch wavevector **p** that we will specify to SCUFF-SCATTER-PERIODIC depends on the incident angle. To identify this dependence, we note that the phase factor in the expression for the incident field is

$$e^{i\mathbf{k}\cdot\mathbf{x}} = e^{ik(x\sin\theta - z\cos\theta)}$$

Hence, for an arbitrary in-plane (two-dimensional) lattice vector $\mathbf{L} = L_x\hat{\boldsymbol{x}} + L_y\hat{\boldsymbol{y}}$, both $\mathbf{E}^{\rm inc,perp}$ and $\mathbf{E}^{\rm inc,par}$ satisfy the condition

$$\mathbf{E}(\mathbf{x}+\mathbf{L}) = e^{ik\sin\theta L_x}\mathbf{E}(\mathbf{x})$$

whereupon we identify the Bloch wavevector as

$$\mathbf{p} = (p_x, p_y) = (k\sin\theta, 0).$$

## Scattered fields

### Reflected fields

We refer to the scattered fields in the region $z > 0$ as the "reflected" fields:

$$\mathbf{E}^{\rm perp,refl}(\mathbf{x}) = -r^{\rm perp}E_0\, e^{i\overline{\mathbf{k}}\cdot\mathbf{x}}\,\hat{\boldsymbol{y}} \tag{1a}$$

$$\mathbf{E}^{\rm par,refl}(\mathbf{x}) = -r^{\rm par}E_0 e^{i\overline{\mathbf{k}}\cdot\mathbf{x}}\,(\cos\theta\,\hat{\boldsymbol{x}} - \sin\theta\,\hat{\boldsymbol{z}}) \tag{1b}$$

where $\overline{\mathbf{k}}$ is the incident wavevector with its $z$ component flipped:

$$\overline{\mathbf{k}} = k\Big(\sin\theta\,\hat{\boldsymbol{x}} + \cos\theta\,\hat{\boldsymbol{z}}\Big).$$

### Transmitted fields

The fields in the region $z < 0$ are the "transmitted" fields:

$$\mathbf{E}^{\rm perp,trans}(\mathbf{x}) = t^{\rm perp}E_0\hat{\boldsymbol{y}}e^{i\mathbf{k}'\cdot\mathbf{x}} \tag{2a}$$

$$\mathbf{E}^{\rm par,trans}(\mathbf{x}) = t^{\rm par}E_0\,(\cos\theta'\hat{\boldsymbol{x}} + \sin\theta'\hat{\boldsymbol{z}})\,e^{i\mathbf{k}'\cdot\mathbf{x}} \tag{2b}$$

The transmitted wavevector is

$$\mathbf{k}' = \sqrt{\epsilon\mu}\cdot k\Big(\sin\theta'\hat{\boldsymbol{x}} - \cos\theta'\hat{\boldsymbol{z}}\Big)$$

with the refraction angle defined by

$$\sin\theta' = \frac{\sin\theta}{\sqrt{\epsilon\mu}}.$$

### Reflection and transmission coefficients

$$t^{\rm perp} = \frac{2Z\cos\theta}{Z\cos\theta + \cos\theta'} \qquad r^{\rm perp} = \frac{Z\cos\theta - \cos\theta'}{Z\cos\theta + \cos\theta'}$$

$$t^{\rm par} = \frac{2Z\cos\theta}{\cos\theta + Z\cos\theta'} \qquad r^{\rm perp} = \frac{\cos\theta - Z\cos\theta'}{\cos\theta + Z\cos\theta'}$$

with

$$Z = \sqrt{\frac{\mu}{\epsilon}}.$$

# 2 Solution using SCUFF-SCATTER-PERIODIC

## 2.1 Creating the surface mesh

## 2.2 Specifying field evaluation points

We will extract the reflection and transmission coefficients by evaluating the scattered fields above and below the plane and comparing against the expected expressions (1) and (2).

More specifically, if $\mathbf{x}$ is an evaluation point *above* the plane, then we can extract the reflection coefficients from the scattered $\mathbf{E}$-field at $\mathbf{x}$ according to

$$r^{\rm perp} = -\frac{\hat{\boldsymbol{y}} \cdot \mathbf{E}^{\rm scat,perp}(\mathbf{x})}{E_0 e^{i\overline{\mathbf{k}}\cdot\mathbf{x}}}$$

$$r^{\rm par} = -\frac{(\cos\theta\,\hat{\boldsymbol{x}} - \sin\theta\,\hat{\boldsymbol{z}}) \cdot \mathbf{E}^{\rm scat,par}(\mathbf{x})}{E_0 e^{i\overline{\mathbf{k}}\cdot\mathbf{x}}},$$

while if $\mathbf{x}$ is an evaluation point *below* the plane then we can extract the transmission coefficients from the scattered $\mathbf{E}$-field at $\mathbf{x}$ according to

$$t^{\rm perp} = \frac{\hat{\boldsymbol{y}} \cdot \mathbf{E}^{\rm scat,perp}(\mathbf{x})}{E_0 e^{i\mathbf{k}'\cdot\mathbf{x}}}$$

$$t^{\rm par} = \frac{(\cos\theta'\,\hat{\boldsymbol{x}} + \sin\theta'\,\hat{\boldsymbol{z}}) \cdot \mathbf{E}^{\rm scat,par}(\mathbf{x})}{E_0 e^{i\mathbf{k}'\cdot\mathbf{x}}}.$$

Here $\mathbf{E}^{\rm scat,perp}$ and $\mathbf{E}^{\rm scat,par}$ are the scattered fields that result from solving the scattering problem with incident fields respectively perpendicular and parallel to the plane of scattering.